

Tutorial: DB and File

Improve the tutorial by ZHU Yueming in 2026.

DB and File Experimental objective

- Analyze that, compare with file, what is the advantage and disadvantage of database .
- Learn how to import data from .sql files.
- Learn how to build the jdbc program according to the source code.
- In this lab, you do not need to understand meaning of those sql statements.

Before your work thinking some question

The following questions are based on processing data by using file system. In this project we only have two .txt files including movies.txt and countries.txt.

1. If we want to add a new movie into movies.txt, what should we do?
2. If we want to get all continent names from countries.txt, what should we do?
3. Suppose that only get the continent names is not enough, and we need continent names with how many countries in each continent, what should we do?
4. More requirements, if we need the information including title of movie, the country name, the continent and the runtime of all movies according to a specific range of runtime, in this case, we have to fetch data from two txt files. What should we do?
5. If we have other requirements, how can we accomplish them?

Import Data

We recommend you import data by using datagrip.

1. Open datagrip and create a new database named `week6` ;

```
create database week6 encoding 'utf8';
```

2. Create a new data source connected to `week6` database:

Data Sources and Drivers

Name: [Create DDL Mapping](#)

Comment:

General Options SSH/SSL Schemas Advanced

Connection type: [default](#) Driver: [PostgreSQL](#) [More Options](#) ▾

Host: Port:

Authentication: ▾

User:

Password: Save: ▾

Database: ▾

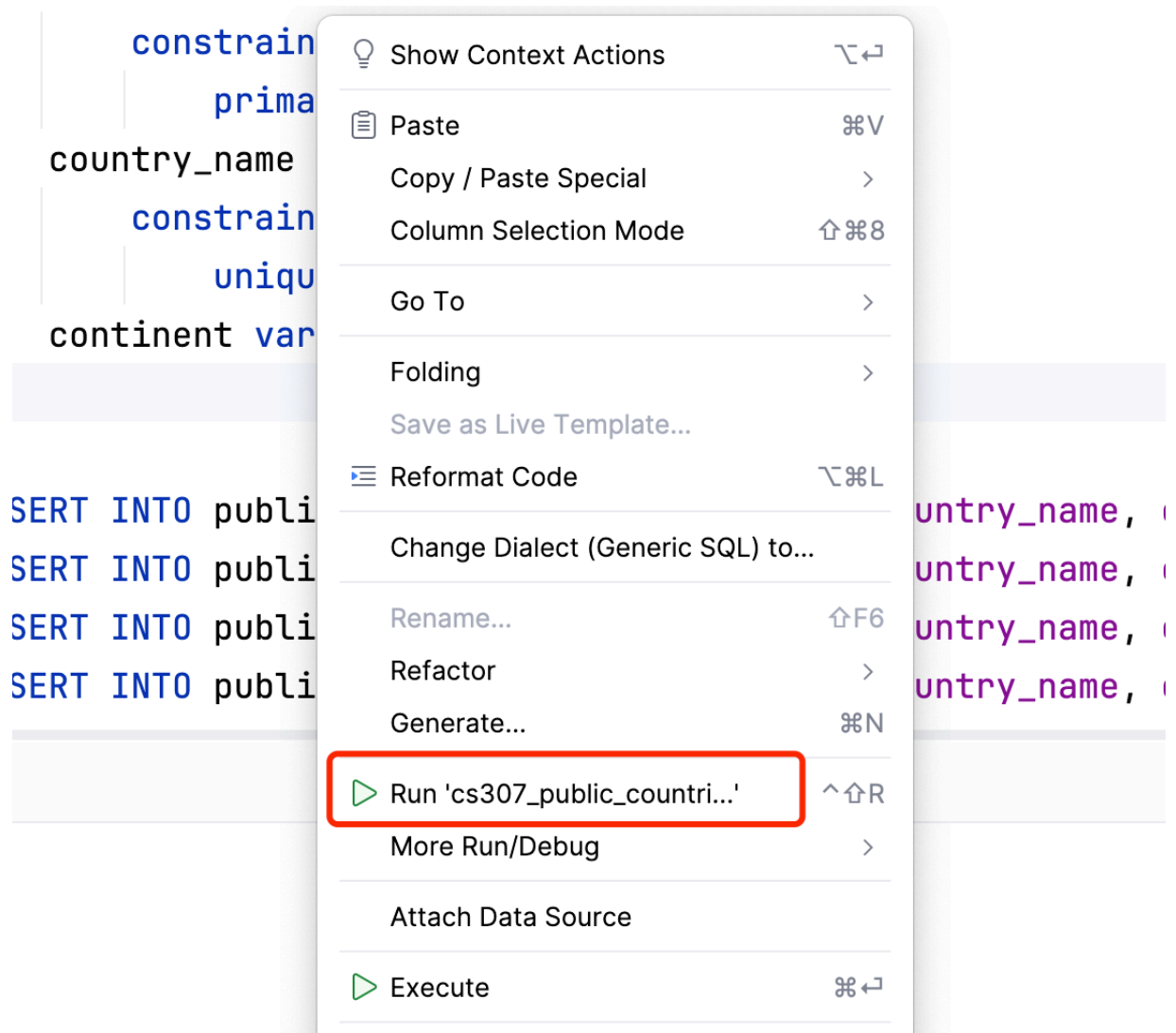
URL: ▾
Overrides settings above

[Test Connection](#) PostgreSQL

3. Drag your `cs307_public_countries.sql` into datagrip

```
1  begin;
2  create table if not exists countries
3  (
4      country_code char(2) not null
5          constraint countries_pkey
6              primary key,
7      country_name varchar(50) not null
8          constraint countries_country_name_key
9              unique,
10     continent varchar(20) not null
11 )
12 ;
13 INSERT INTO public.countries (country_code, country_name, continent) VALUES
14 INSERT INTO public.countries (country_code, country_name, continent) VALUES
15 INSERT INTO public.countries (country_code, country_name, continent) VALUES
16 INSERT INTO public.countries (country_code, country_name, continent) VALUES
```

4. Right click the file, and find `Run 'cs307_public_countires.sql'`



5. Add the datasource, and then click Run.

Edit Configuration

Name: ☐ Allow multiple instances ☐ Store as project file

Target data source / schema:

week6@114.132.77.229 / week6.public

Execute: ☐ Script text ☒ Script files

rse.nosync/26 Spring DB/24-github-file/Part06. DB and File/data/cs307_public_countries.sql ✓

▼ Before launch

There are no tasks to run before launch

☒ Show this page ☒ Activate tool window ☐ Focus tool window

6. Similar way to import `cs307_public_movies.sql`

Test your import

Execute:

```
select count(*) from movies;
```

result: 9538

```
select count(*) from countries;
```

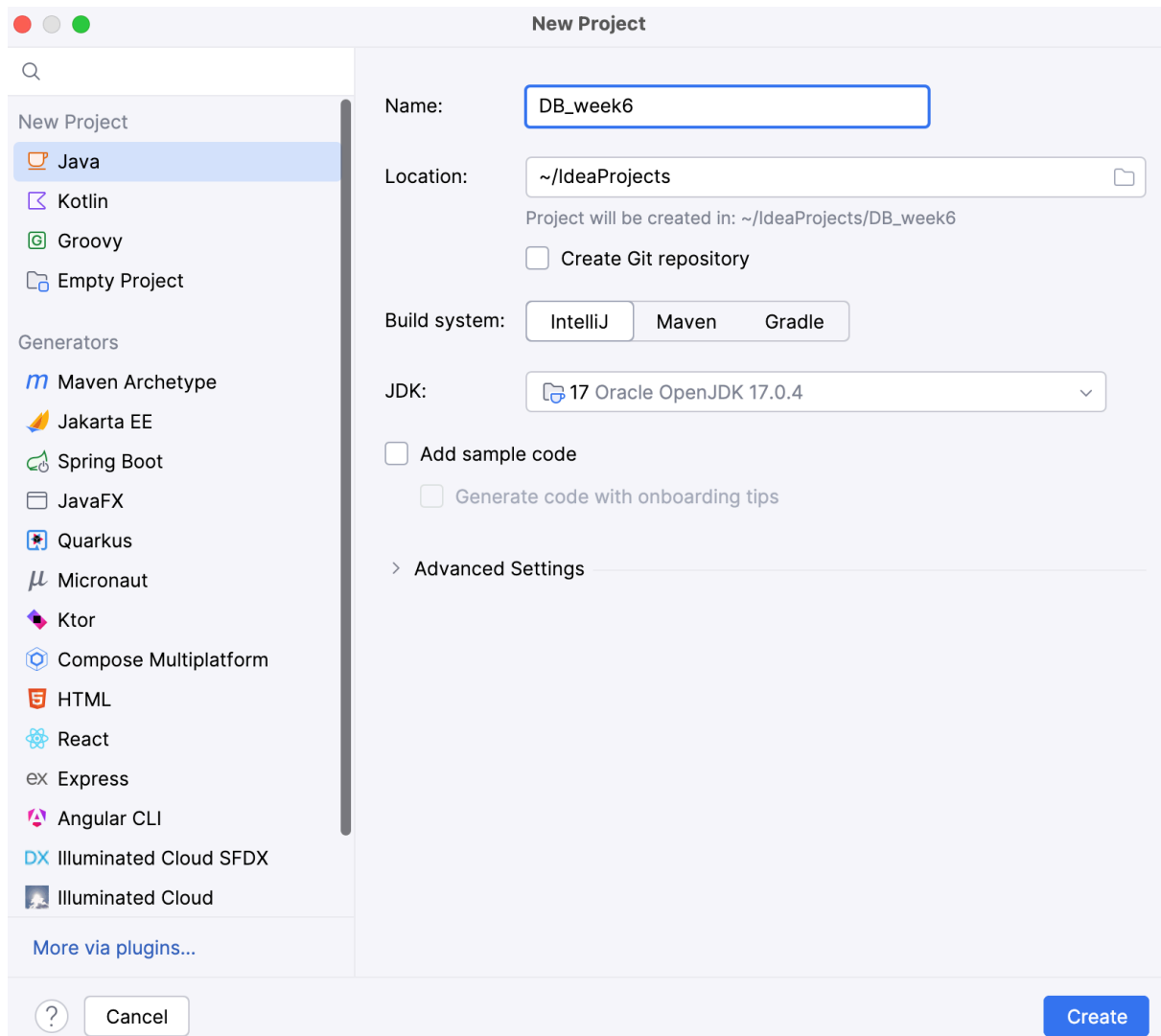
result: 185

Build Your Project

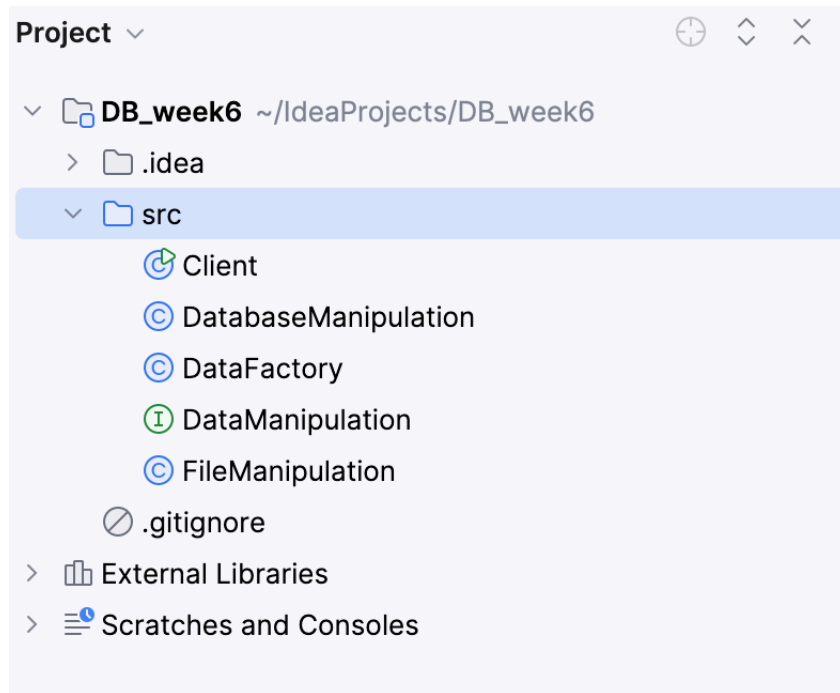
We recommend you build project by using **IntelliJ IDEA**. In this tutorial, we use IntelliJ IDEA as IDE, and you can also do it by other way.

1. File

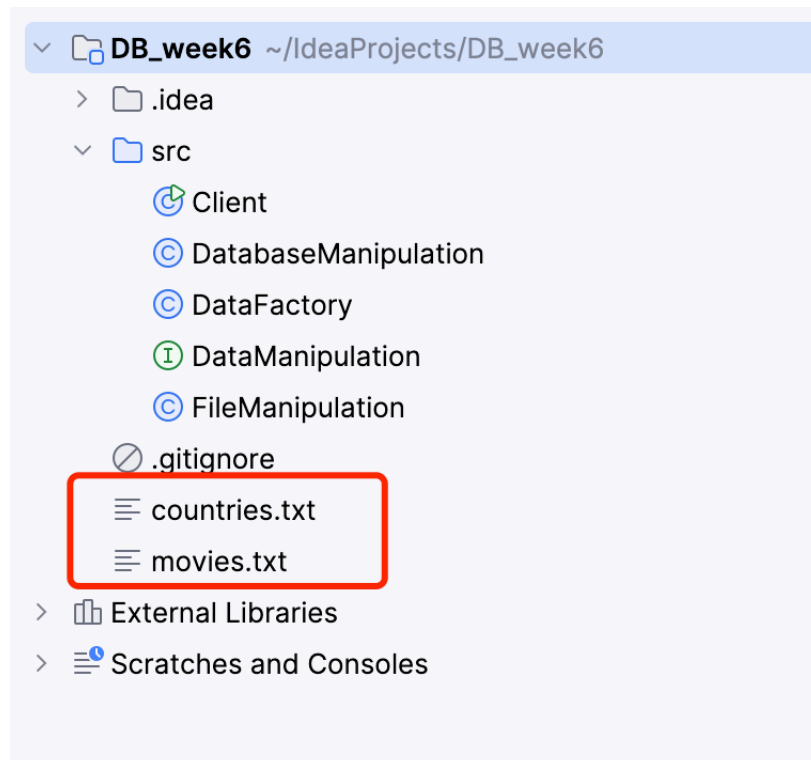
1. Create a new project named `db_week6`.



2. Drag five .java files into **src** (generated sources root) folder.



3. Drag two .txt files into your project folder.

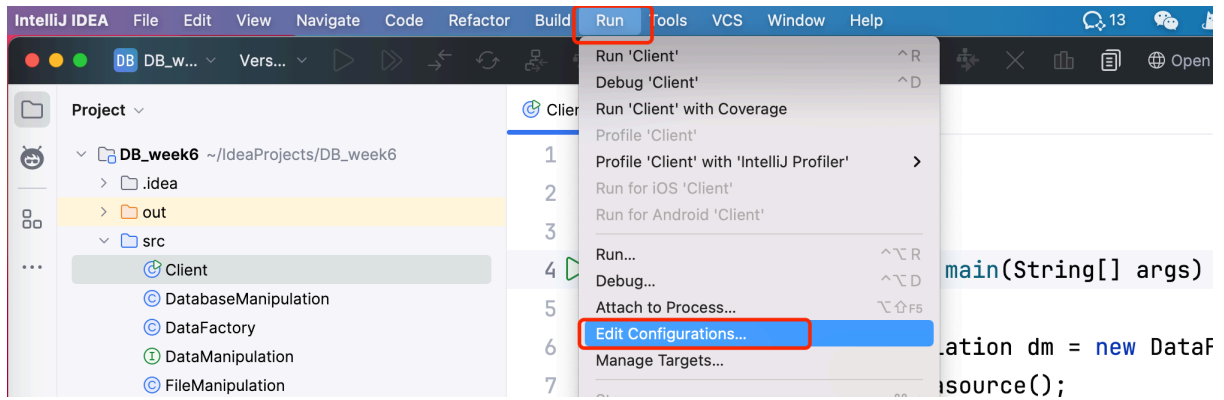


4. Run `Client.java`

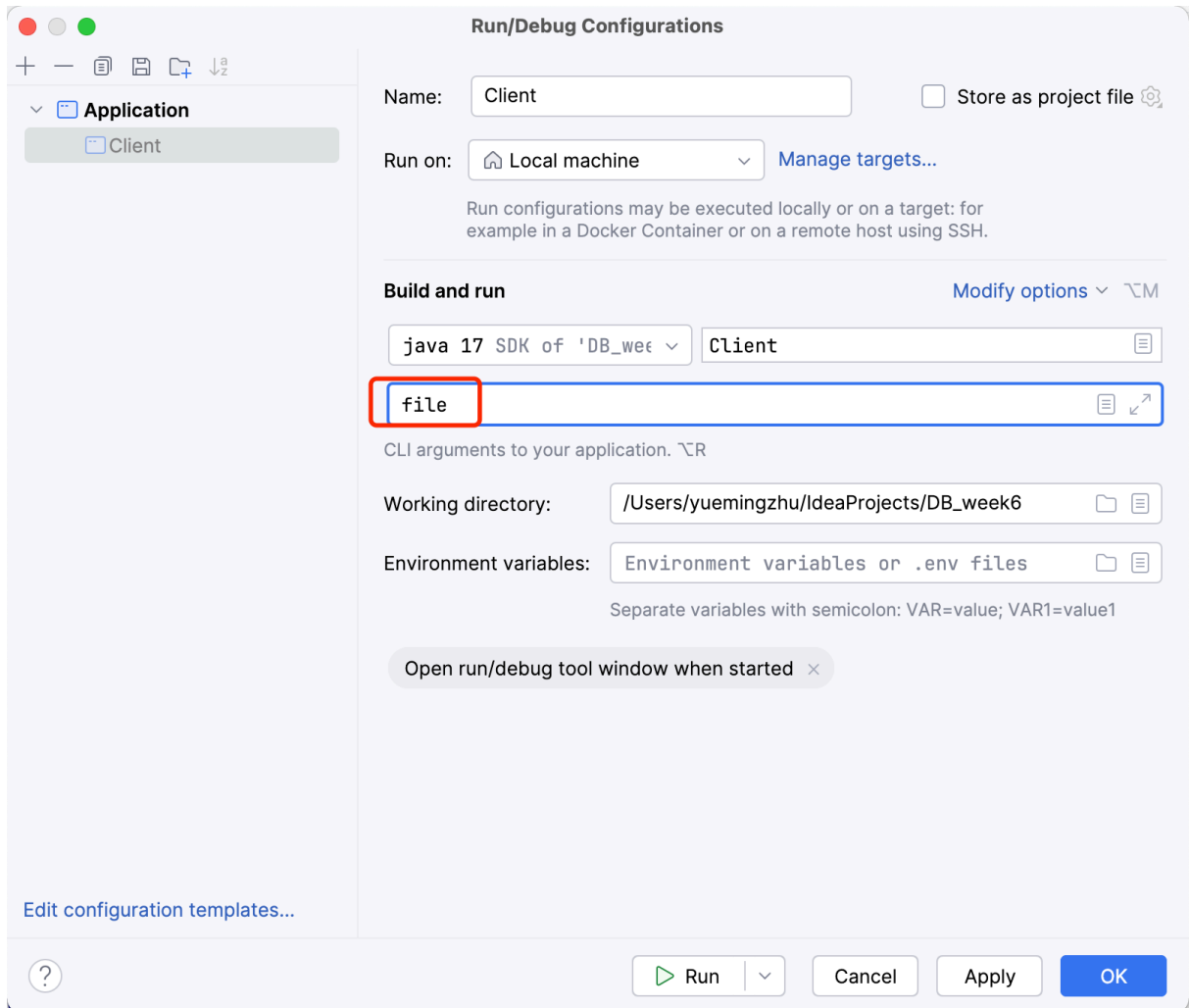
It will meet an exception as follows. Exception in thread "main"
java.lang.ArrayIndexOutOfBoundsException:.....

5. Setting Configurations

Run -> Edit Configurations -> Configuration -> Program arguments



Add arguments `file`



6. Run again the result would be

```

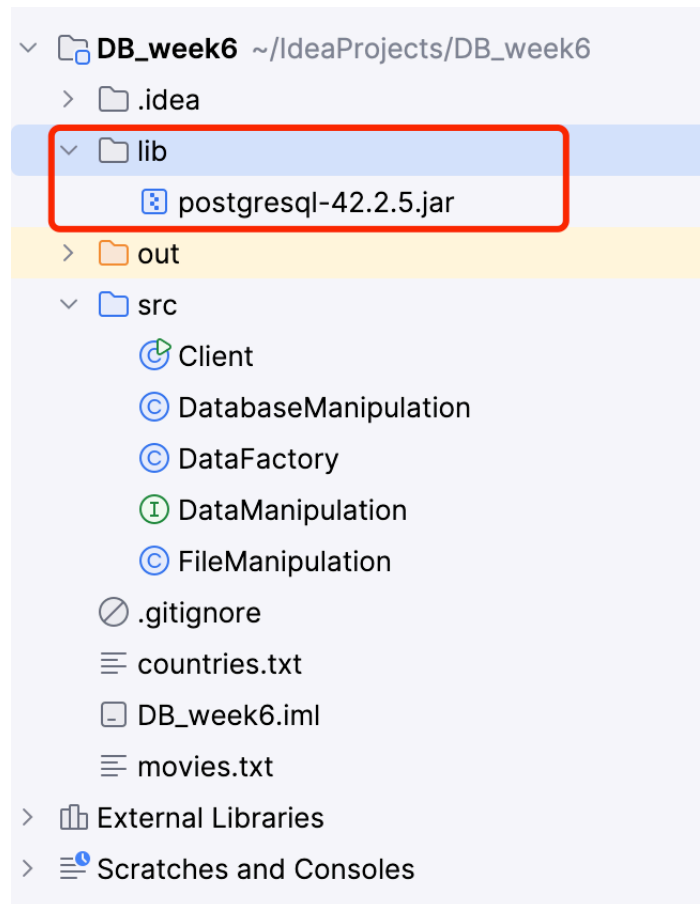
ASIA
EUROPE
AFRICA
AMERICA
OCEANIA

EUROPE  40
AFRICA  52
.....

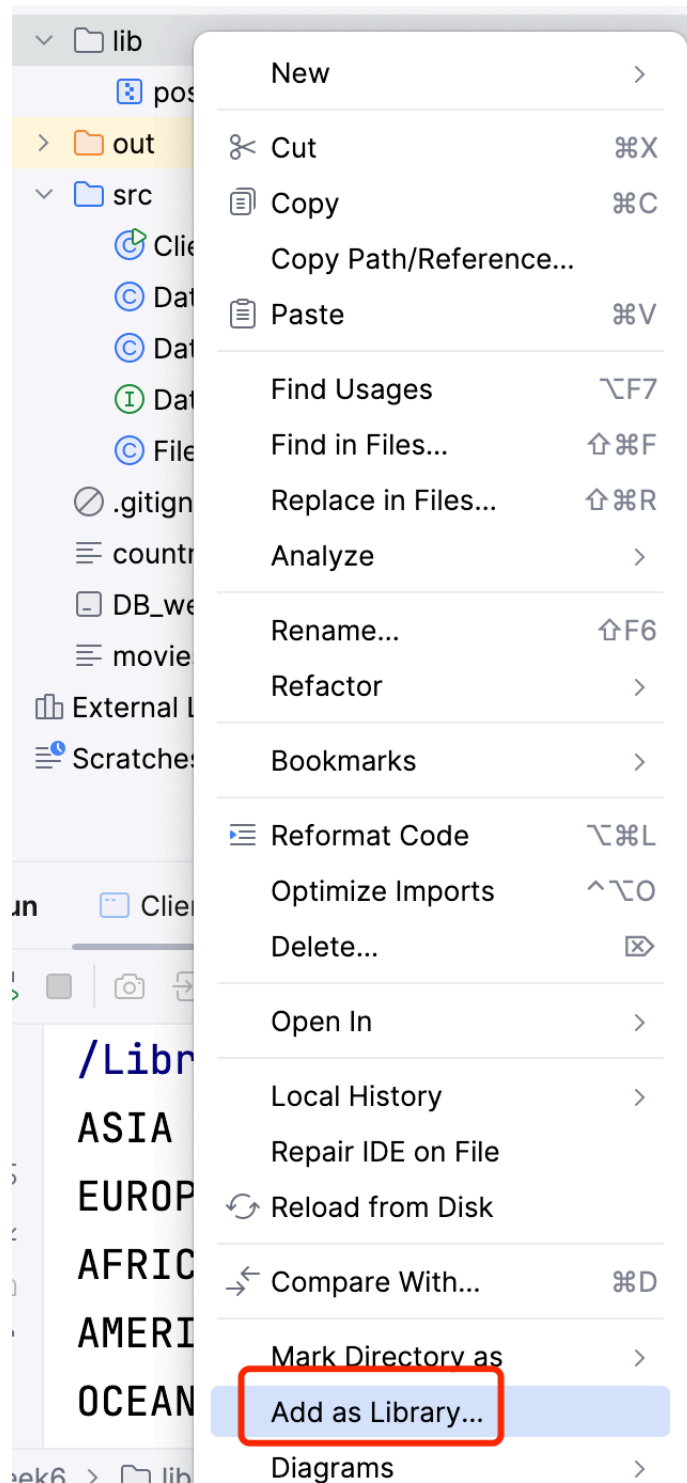
```

2. Database

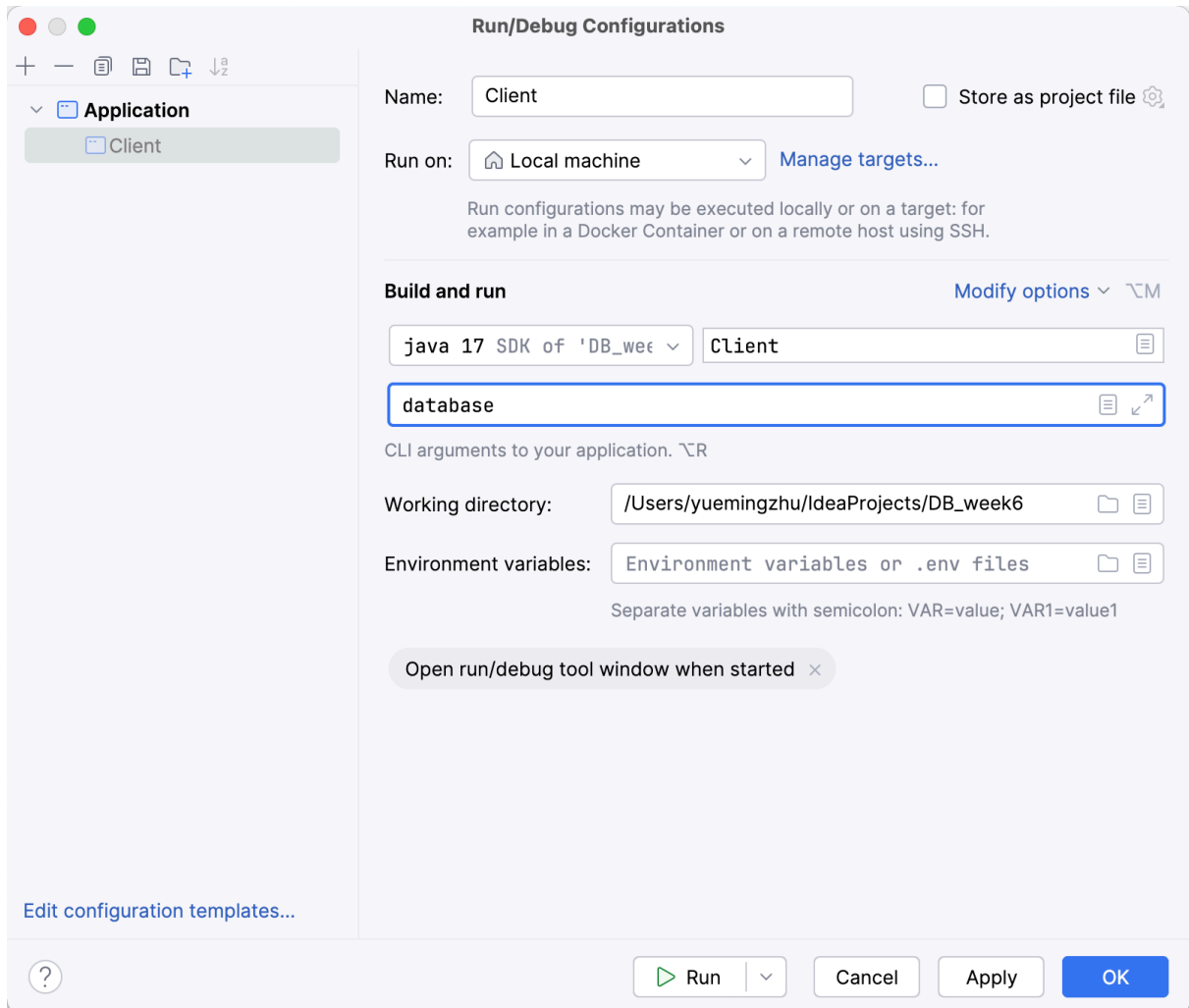
1. Create a document named `lib` in project folder. Then import postgres Driver into your project.



2. Add the lib as library.



3. Change Program argument to **database**



4. Change the data in `DatabaseManipulation.java` of your database.

```
DatabaseManipulation.java x

import java.sql.*;

public class DatabaseManipulation implements DataMa
    private Connection con = null; 8 usages
    private ResultSet resultSet; 13 usages

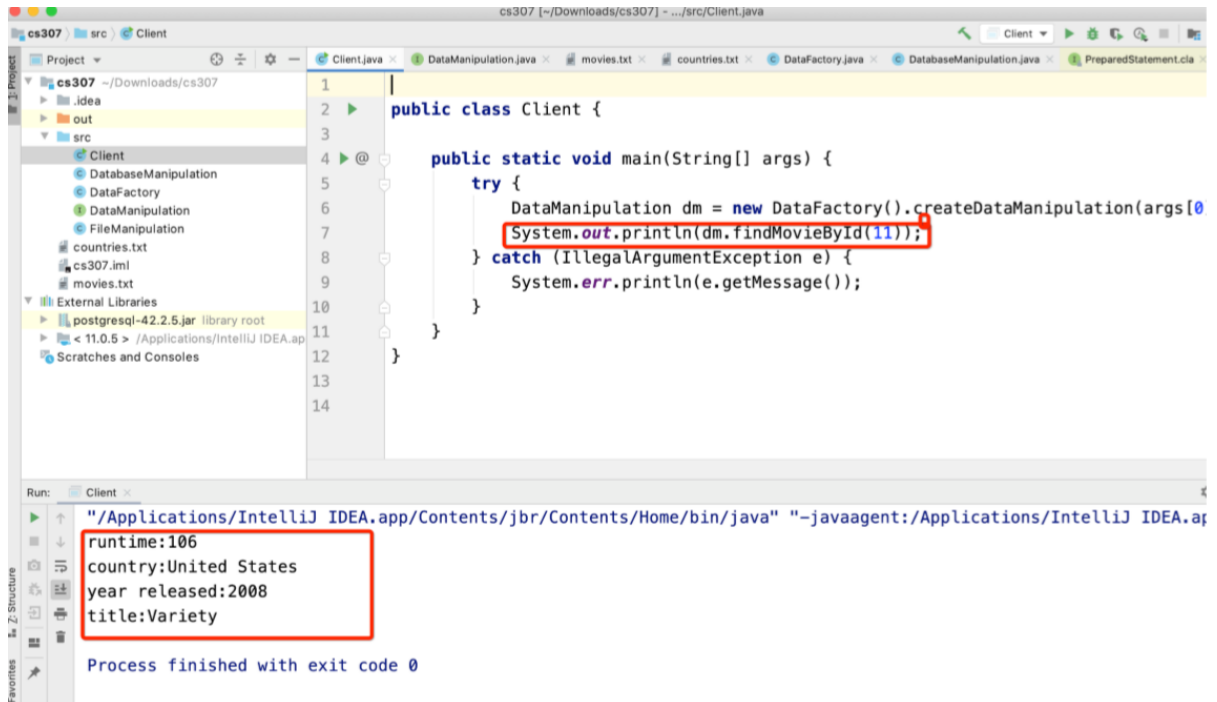
    private String host = "localhost"; 1 usage
    private String dbname = "week6"; 1 usage
    private String user = "checker"; 1 usage
    private String pwd = "123456"; 1 usage
    private String port = "5432"; 1 usage
```

Exercise

- Try the query

```
select m.title, c.country_name, m.year_released, m.runtime from movies m
join countries c
on m.country = c.country_code
where m.movieid = 5;
```

- Add a method named findMovieById(int id)
- According to the code given to you, please implement method findMovieById in DatabaseManipulation. The result would be:



Tips: You can use executeQuery() method to execute the query in prepare statement, and the result would stored in resultSet

```
resultSet = preparedStatement.executeQuery();
```

If you want to find the first selecting result in resultSet, you can do:

```
resultSet.next();
```